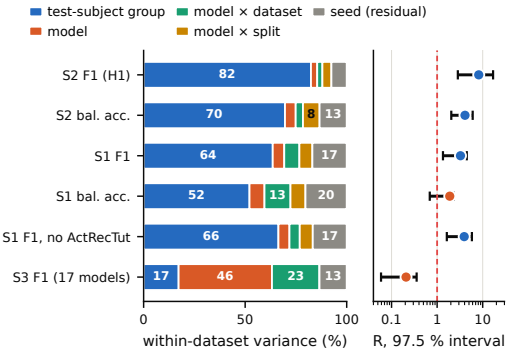
Who Is Tested or Which Network? A Pre-Registered Variance Decomposition and Error Budget for Person-Disjoint Wearable Activity Recognition

[Authors to be added by the authors]

[Affiliations to be added by the authors]

Abstract—Wearable activity-recognition networks are ranked by macro-F1 on unseen people, yet the network's share of that score has not been separated from that of who is tested. We decompose stored runs of one person-disjoint protocol (WHAR Arena v1) with a crossed mixed model. A pre-registered test on 7 held-out datasets (828 runs) found that the test-subject group carries 8.19 times (97.5 % interval 2.85–16.90) the variance of four shared-skeleton networks. Among the benchmark's published networks the ratio stays near or above 1 for the best and falls to 0.21 across all 17. The components give an error budget: paired over 10 test groups and 3 seeds, a comparison detects 4.1 F1 points; unpaired on one split, 47.9. Validation windows drawn from the training subjects rank networks poorly (median Spearman +0.18; +0.80 between test halves). Comparisons should be paired, sized by test subjects and validated on held-out subjects.



Index Terms—Wearable sensors, inertial sensors, human activity recognition, person-disjoint evaluation, variance components, benchmark evaluation.

I. INTRODUCTION

Wearable inertial sensing systems for human activity recognition (HAR) are designed and chosen on one number: macro-F1 on subjects held out from training [1]. Evaluating on unseen people removes the optimism of random cross-validation [2], and how people are partitioned between training and test data changes the reported accuracy [3]. People move differently [4] and devices sense differently [5], so a test set of a few subjects carries its own variation. That variation is large: datasets differ in how much their subjects vary [6], and under leave-one-subject-out evaluation the spread over subjects makes confidence intervals too wide to compare methods [7], as small test samples do in other fields [8]. WHAR Arena [9] fixed one person-disjoint protocol across 30 datasets and found the state of the art distributed over architectures rather than held by one; it reports one training run per split, so it cannot separate the effect of who is tested from the training seed. Machine-learning benchmarks depend on several such sources of variation [10]. What is missing for wearable HAR is their separation: how much of the variance of a reported score belongs to the test subjects, to the network, to their interaction and to the seed, and therefore which of them a comparison must control. Earlier work narrowed the intervals by estimating them from the training samples instead of the subjects [7]; we keep the subject variance and separate it from the network terms, which shows what pairing by split removes and what it leaves.

This letter makes four contributions.

- A crossed variance-components decomposition of person-disjoint macro-F1 into the test-subject group, the network, the network × dataset and network × split interactions and the training seed, with a pre-registered test on held-out datasets: for networks that differ in single components, the test-subject group carries most of the within-dataset variance.
- Evidence that this ratio is a property of the networks compared. Among the benchmark's published networks it stays near or above 1 for the best ones and reverses once weaker networks enter, so the letter states where "who is tested" outweighs "which network", and where it does not.
- An error budget for comparisons, derived from the components: the noise of paired and unpaired comparisons, the smallest detectable difference and the reliability, by number of test-subject groups and seeds.
- Evidence that the protocol's validation set, drawn window by window from the training subjects, overstates the person-disjoint score and ranks networks far worse than the test ranks them against itself, so that early stopping and model selection in the benchmark rest on a weak signal.

Exploratory analyses (the spread of test groups and networks, sensing properties of the datasets, what the temporal memories learned, and how far the test subjects lie from the training subjects in the embedding space) support these results; all are labelled exploratory.

II. METHODS

A. Datasets and sensors

The 18 datasets of the stored runs (Table 1, Fig. 1) were recorded with smartphones, body-worn inertial nodes, a smartwatch, smart glasses and a helmet. Every dataset has accelerometers; most add gyroscopes, several magnetometers, and single datasets a barometer, an ambient-light sensor, stretch sensors, ECG or EEG. They span 3 to 133 channels, 2 to 36 subjects, 3 to 27 classes and 20 to 250 Hz, with 232,254 windows in all. Devices and modalities are quoted from the benchmark's dataset table [9]; channel and class counts, sampling rates and window counts are read from the stored runs, and subject counts from the harness's splits (for USC-HAD and MotionSense, whose caches we do not hold, from the dataset configurations).

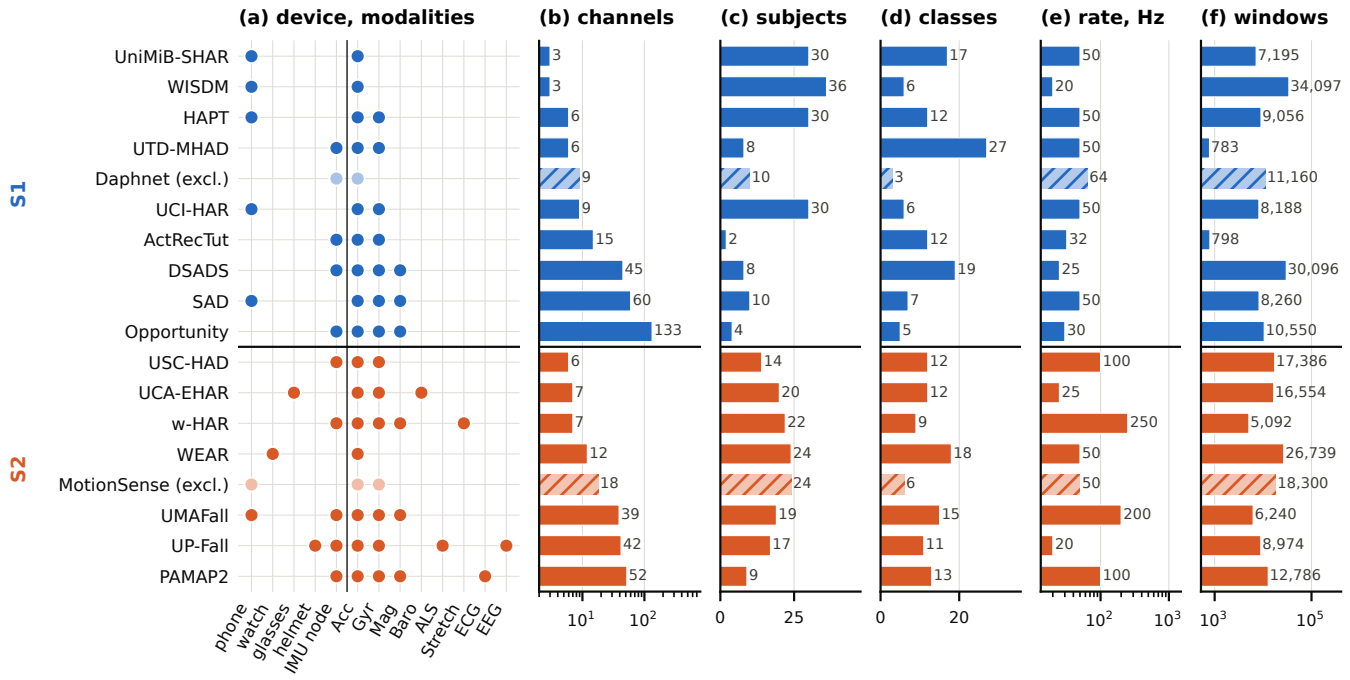


Fig. 1. Sensor configuration of the datasets of S1 (development) and S2 (held-out). (a) Devices and sensor modalities, quoted from the benchmark's dataset table [9]; (b)–(f) channels, subjects, classes, sampling rate and number of 3-s windows (logarithmic axes in (b), (e) and (f)), read from the stored runs and the harness's splits. Faded and hatched: excluded from the variance analyses by the pre-registered inclusion rule.

TABLE 1. Datasets of the stored runs

Dataset	Set	Device	Modalities	C	Subj.	Cl.	Hz	Windows	Splits
UniMiB-SHAR	S1	phone	Acc	3	30	17	50	7,195	10
WISDM	S1	phone	Acc	3	36	6	20	34,097	10
HAPT	S1	phone	Acc, Gyr	6	30	12	50	9,056	10
UTD-MHAD	S1	node	Acc, Gyr	6	8	27	50	783	8
Daphnet†	S1	node	Acc	9	10	3	64	11,160	10
UCI-HAR	S1	phone	Acc, Gyr	9	30	6	50	8,188	10
ActRecTut	S1	node	Acc, Gyr	15	2	12	32	798	2
DSADS	S1	node	Acc, Gyr, Mag	45	8	19	25	30,096	8
SAD	S1	phone	Acc, Gyr, Mag	60	10	7	50	8,260	10
Opportunity	S1	node	Acc, Gyr, Mag	133	4	5	30	10,550	4
USC-HAD	S2	node	Acc, Gyr	6	14	12	100	17,386	10
UCA-EHAR	S2	glasses	Acc, Gyr, Baro	7	20	12	25	16,554	10
w-HAR	S2	node	Acc, Gyr, Mag, Stretch	7	22	9	250	5,092	10
WEAR	S2	watch	Acc	12	24	18	50	26,739	10
MotionSense†	S2	phone	Acc, Gyr	18	24	6	50	18,300	10
UMAFall	S2	node, phone	Acc, Gyr, Mag	39	19	15	200	6,240	10
UP-Fall	S2	node, helmet	Acc, Gyr, ALS, EEG	42	17	11	20	8,974	10
PAMAP2	S2	node	Acc, Gyr, Mag, ECG	52	9	13	100	12,786	9

C: channels; Subj.: subjects; Cl.: classes; Hz: sampling rate; Windows: 3-s windows, each in one test group; Splits: test-subject groups. †Excluded from the variance analyses by the pre-registered inclusion rule.

B. Protocol and stored runs

All runs follow WHAR Arena v1 with its paper-era code [9], [11]: 3-s windows with 50 % overlap at the native sampling rate; $k = 10$ subject groups, each split testing on one group (1 to 4 subjects) that never appears in training or validation; validation from 20 % of the training subjects' windows, drawn window by window, so that a validation window can share half its samples with a training window; a z-score fitted on the training split; macro-F1 over all classes, with classes absent from a test split counted as 0. Apart from a label-permutation control, no network was trained for this letter: we analyse stored runs and their checkpoints. Three sets enter (Table 3): S1, ten networks that share one skeleton and differ in single components (with and without temporal memory, e.g. S4D [12], [13] and Mamba [14], or window statistics), on ten development datasets with three seeds; S2, four of these networks (RecurrenceNet, its two variants with a side path, and S4D) on eight datasets not used to find the effect, with three seeds; S3, the benchmark's 17 published models (deep networks such as DeepConvLSTM [15], [16], and Random

TABLE 2. The ten networks

Network	Memory	Side path	Sets	$D \times N$	Emb.	Params (ref.)	Range over datasets
RecurrenceNet	recurrence	none	S1, S2	32×3	32	24,675	23,886–28,013
S4D	S4D (diagonal SSM)	none	S1, S2	64×2	64	27,675	26,118–34,373
HysteresisNet	play operators	none	S1	32×3	32	23,579	22,790–26,917
QuantileNet	quantile trackers	none	S1	32×3	32	20,763	19,974–24,101
Mamba	Mamba (S6)	none	S1	32×2	32	21,147	20,358–24,485
No memory	none (play width 0)	none	S1	32×3	32	19,995	19,206–23,333
RN + stats	recurrence	channel stats	S1, S2	32×3	64	26,451	24,582–46,357
RN + laws	recurrence	law invariants	S1, S2	32×3	64	27,553	25,316–49,437
Laws (no mem.)	none (plain block)	law invariants	S1	48×3	80	19,033	16,412–42,597
Stats (no mem.)	none (plain block)	channel stats	S1	48×3	80	17,931	15,678–39,517

Emb.: embedding dimension. Params (ref.): with the reference input of gate G1; range over the datasets of S1 and S2.

Forest [17], SVM and k-NN on tsfresh features [18]) re-run with the same code on four datasets, one run per split. A pre-registered inclusion rule removed datasets whose stored runs disagree on split sizes, because their split indices do not name the same subjects across runs (one dataset each in S1 and S2). Fig. 2 summarises the design, the runs analysed and how subjects are assigned to test groups.

C. Networks

The ten networks of S1 and S2 share one skeleton (Fig. 3, Table 2). A linear encoder maps the C channels of each sample to D features, N residual blocks process the sequence, and the features are averaged over the window, joined in four networks by a window-level side path, normalised (LayerNorm) and classified by a linear head. The vector the head receives is the network's embedding (Section II-F). In a residual block, a LayerNorm and a linear map give per-channel signals u that feed a memory; u and the memory's outputs are concatenated, mixed linearly and passed through a GELU into the residual sum. RecurrenceNet's memory keeps soft rows of a recurrence plot [19] over ten lags: whether a learned two-dimensional projection of the signal is back where it was τ samples earlier. HysteresisNet keeps four play operators per channel, whose output moves only when the input leaves a band of half-width r around it [20] (here with input-dependent widths and a clock channel). QuantileNet keeps four online quantile trackers per channel, stochastic approximation steps [21] that settle on a running quantile of level τ. The no-memory control sets every play width to zero, which makes each operator the identity, and the two networks without memory use plain blocks. S4D [13] and Mamba [14] use their own blocks, a diagonal state-space convolution and a selective scan. The side paths read per-channel window statistics (mean, standard deviation, minimum and maximum, the kind of features behind the Random Forest) or invariants of each three-axis sensor's temporal operator and of the couplings between sensors ("laws"). With the reference input of gate G1 (that of UTD-MHAD), the networks have 17,931 to 27,675 parameters, and on the datasets of S1 and S2 15,678 to 49,437.

D. Variance model and pre-registered test

For run (m, d, s, r) of network m on split s of dataset d with seed r,

$$y_{mdsr} = \mu_d + P_{ds} + M_m + (MD)_{md} + (MP)_{mds} + e_{mdsr}, \quad (1)$$

with μ_d fixed and the other terms independent zero-mean random effects, fitted by REML [22] in statsmodels [23]. The estimand is $R = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_{MP}^2)$, the variance of the test-subject group over all network-related variance; π_P is σ_P^2's share of the within-dataset variance. A split changes the test subjects and, with them, the training and validation subjects, so P is the effect of the subject partition. On the datasets of S2, with nine or ten groups, any two splits share most of their training subjects, so P is mainly the effect of the test group. Intervals are percentile 97.5 % intervals from 1000 cluster-bootstrap resamples that draw splits within datasets [24]; they treat the networks as fixed. Before any variance analysis of S2, we registered H1 ($R > 1$ on S2), with success if the interval's lower bound exceeds 1 and failure if $R \leq 1$, sealed the analysis code by hash, and logged S2's single read. S2's per-dataset means and its paired network differences were known from an earlier study; its variance components were not. For S3 (one run per split) the interaction and the seed cannot be separated, so the conservative $R' = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_e^2)$ is reported.

E. Exploratory analyses

None of the following analyses was pre-registered; four were added after an internal review, and are marked (r).

Network population (r). We refit S3's model to sub-populations of its networks: the 14 deep networks, and the k best networks for k from 3 to 17, ranked by their mean published macro-F1 on the 26 Arena datasets outside S3 [9], so that the ranking reads no score of the runs analysed. Each point estimate keeps the better of the statsmodels fit and an independent REML implementation; intervals use the bootstrap of H1.

Error budget (r). Under (1), the difference between two networks on one dataset, averaged over k test-subject groups and n seeds, has a true part with variance $2(\sigma_M^2 + \sigma_{MD}^2)$ over pairs of networks and a noise part with variance $2(\sigma_{MP}^2 + \sigma_e^2/n)/k$ when both networks run on the same splits (paired), and $2(\sigma_P^2 + \sigma_{MP}^2 + \sigma_e^2/n)/k$ when they run on different test subjects (unpaired, e.g. two studies that each report one hold-out split). From the point estimates we derive the noise SD, the smallest true difference a two-sided test at 5 % detects with 80 % power, and the reliability (true over true plus noise variance). k beyond 10 means more groups of the same size, that is, more recorded subjects.

Validation against the test (r). For each dataset of S1 and S2 whose runs all recorded a validation score, we rank-correlate the networks' validation and test means, and compare this with how well the test ranking agrees with itself: test means of seed 0 against seeds 1–2 (same splits), and of the even against the odd splits (other test subjects).

Further analyses. Component contrasts on S1 pair two networks by split, average the seeds, and summarise over datasets with a t-interval and a Wilcoxon test [25], [26]. Over the benchmark's 30 datasets we regress the published single-run results on standardised dataset properties (ordinary least squares; four properties, four outcomes); these associations are correlational, and we report Holm-adjusted p-values over the 16 coefficient tests [27] (r). Within each dataset of S1 and S2 we compare the spread of the split means (over networks and seeds) with that of the network means (over splits and seeds). From the stored checkpoints of eight S1 datasets we read what the memories learned and compare it with the same networks freshly initialised: S4D's mode time constants and decay rates, the play widths, the quantile levels and steps, and the recurrence radii.

F. Embedding analysis

This analysis is exploratory and uses inference only. For the seed-0 checkpoint of every split of the 11 datasets whose caches on our

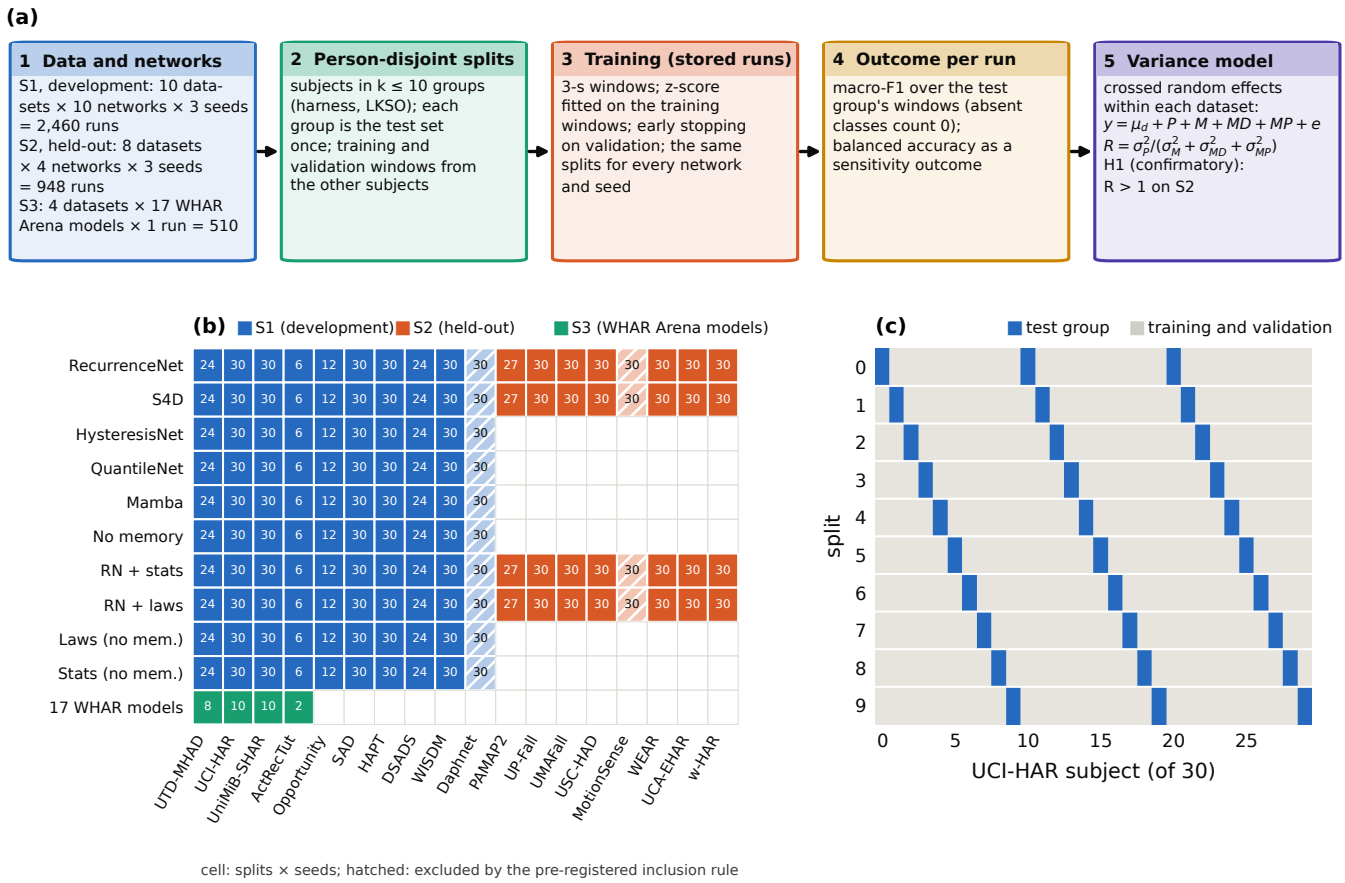


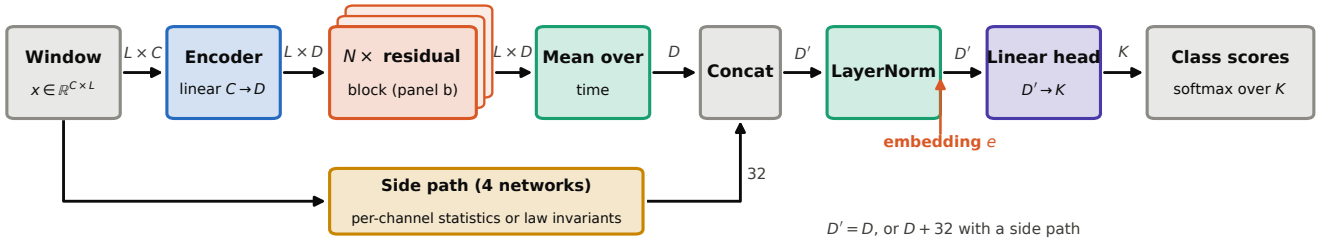
Fig. 2. Study design. (a) From the stored runs to the variance decomposition. (b) The runs: networks (rows) by datasets (columns), coloured by run set; each cell gives splits × seeds, and hatched cells are excluded by the inclusion rule. (c) The person-disjoint test groups of UCI-HAR: each split tests on three subjects that never enter its training or validation data.

cluster reproduce the stored runs (eight of S1 with the ten networks, three of S2 with four), the harness rebuilds the split and the network embeds every training and test window; the regenerated macro-F1 equals the stored one in 686 of 754 cases (largest difference 0.20 points). Three quantities describe each network and split. First, the class-conditional shift of the test group [28]: for every class present in both, the distance between the test and the training class means, in units of the training class's root-mean-square spread per dimension, averaged over classes. The same shift is computed in a sensor-feature space that no network shapes (per-channel mean, standard deviation, minimum and maximum of the normalised window). Second, the subject information in the training embeddings: the balanced accuracy of a linear probe [29] (logistic regression) that predicts the training subject, in five folds grouped by recording session so that overlapping windows never straddle folds, on at most 4,000 windows, against chance (one over the number of training subjects). Third, the same probe for the activity. Within each dataset we rank-correlate the test group's shift, averaged over networks, with the split's macro-F1 (Spearman), and we pool the within-dataset ranks over datasets with more than three splits. One case, fixed before the analysis (UCI-HAR, S4D, split 0), is mapped to two dimensions with t-SNE [30].

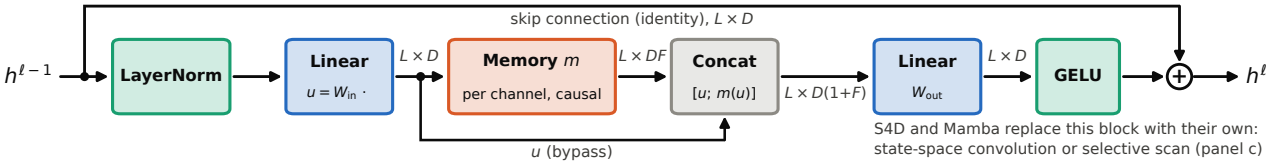
G. Verification

For verification, the benchmark's Random Forest, re-run with our code, reproduces its published macro-F1 on four datasets (e.g. 84.69 against 84.69 on UCI-HAR); test predictions were regenerated from 408 checkpoints, 373 of which reproduced the stored metrics exactly and all within two test windows; an independent REML implementation and negative controls (e.g. permuting network labels within cells) confirmed the analysis; and S4D retrained on permuted training labels (6 runs on UP-Fall and UCA-EHAR) fell to chance, with a test macro-F1 of 4.9 against 8.7 by chance and 54.5 with the true labels. Of the 828 confirmatory runs, the gate regenerated 360 on the caches the stored runs used, within that tolerance. Beyond the gate, every analysed run with a checkpoint was re-executed (2,982 of 2,988): 2,185 reproduce their records exactly, 2,668 within two test windows, and none differs by more than 3.25 F1 points. The residual differences have one source. Several parsers number activities and subjects, and order recordings, in the order the file system lists them, and many stored runs were built on cloud machines whose listing order cannot be recreated. For DSADS we recovered that order from the checkpoints themselves, since each network is near-perfect on its own training subjects, which identifies its output labels and its test subjects, and rebuilt the cache accordingly; for USC-HAD and three datasets whose caches we rebuilt, the test sets are identical and only the draw of validation windows, and with it the train-only z-score,

(a) Shared skeleton of the ten networks



(b) Residual block $\ell = 1, \ldots, N$ (recurrence, play, quantile and plain blocks)



(c) Memories: update per channel and time step t (input u_t)

Memory	Update rule	State per channel	Networks
Recurrence plot	$r_t(\tau) = \sigma((\varepsilon - \lVert z_t - z_{t-\tau} \rVert)/T)$ and $r_t(\tau)\, r_{t-1}(\tau)$, z_t: 8 learned 2-D projections of u_t; 10 lags $\tau \le 48$	48-sample buffer	RecurrenceNet, RN + stats, RN + laws
Play operator	$y_t = \min(\max(y_{t-1}, u_t - r), u_t + r)$, 4 widths r (HysteresisNet: r depends on u_t, plus a clock channel)	4 values y	HysteresisNet; No memory ($r = 0$)
Quantile tracker	$q_t = q_{t-1} + \eta(\tau - \mathbf{1}[u_t < q_{t-1}])$, 4 levels τ	4 values q	QuantileNet
Diagonal SSM	$s_t = \bar{A}\, s_{t-1} + \bar{B}\, u_t$, $y_t = \mathrm{Re}(C\, s_t) + D u_t$; FFT convolution	16 complex modes	S4D
Selective scan	$s_t = \bar{A}_t s_{t-1} + \bar{B}_t u_t$, $y_t = C_t s_t$; Δ_t, B_t, C_t computed from u_t	16 states	Mamba
None	plain block: the memory and the concatenation are absent	none	Laws, Stats (no mem.)

Fig. 3. The networks of S1 and S2 as a dataflow diagram. (a) The shared skeleton, with the tensor shapes on its edges (C channels, L samples, width D, K classes); the embedding analysed in Section II-F is the head's input. (b) The residual block with its skip connection; the memory returns F features per channel. (c) Each memory's update per channel and time step, the state it keeps, and the networks that use it (sizes in Table 2).

TABLE 3. Variance components (F1 points^2) and R [97.5 % interval]

Analysis	n	σ^2_P	σ^2_M	σ^2_{MD}	σ^2_{MP}	σ^2_e	R [97.5 % int.]
S2 F1 (H1)	828	127.7	4.8	4.0	6.9	11.7	8.19 [2.85, 16.90]
S2 bal. acc.	828	86.3	6.5	4.4	10.1	16.6	4.09 [2.05, 6.03]
S1 F1	2160	66.2	5.9	7.8	6.7	17.6	3.26 [1.34, 4.47]
S1 bal. acc.	2160	39.2	5.5	9.6	5.5	15.3	1.89 [0.69, 2.29]
S1 F1, no ActRecTut	2100	67.4	5.5	5.1	6.5	17.0	3.93 [1.63, 5.80]
S3 F1 (17 models)	510	37.5	100.2	51.0	–	29.2	0.21 [0.06, 0.36]

differs. The 6 runs without a checkpoint were retrained to obtain one; 5 of them reproduce their record's macro-F1 exactly, and the stored records remain the ones analysed.

III. RESULTS

A. Confirmatory test

On S2 the test-subject group dominates (Fig. 4, Table 3): σ^2_P = 127.7 (F1 points^2) against σ^2_M = 4.8, σ^2_{MD} = 4.0 and σ^2_{MP} = 6.9, with the seed at σ^2_e = 11.7. Hence R = 8.19 [2.85, 16.90] and π_P = 82 % [64, 89]: the pre-registered decision is *success*, and the secondary criterion (π_P lower bound ≥ 40 %) is met.

B. Sensitivity analyses

Balanced accuracy, which averages recall over the classes present and so has no per-split ceiling, gives R = 4.09 [2.05, 6.03] on S2: ceilings enlarge the subject share but do not create it. On the development set S1 the ordering is the same with macro-F1 (R = 3.26 [1.34, 4.47]), also without its two-subject dataset ActRecTut (R = 3.93); with balanced accuracy the point estimate stays above 1 but the interval does not exclude it (R = 1.89 [0.69, 2.29]). Across the benchmark's 17 published architectures it reverses: R' = 0.21 [0.06, 0.36], with σ^2_M = 100.2 against σ^2_P = 37.5. These architectures range from k-NN to attention networks; Section III-C shows which of them cause the reversal.

C. Which networks are compared

This analysis is exploratory (r). The reversal on S3 comes from the networks at the bottom of the benchmark (Fig. 5a). Ranked by their published macro-F1 on other datasets, the best k networks keep R' above 1 up to k = 12: for the best five R' = 1.86 [0.36, 3.94], for the best 12 R' = 1.26 [0.32, 2.25], although no interval excludes 1 (0 of 15 populations). Adding the three weakest deep networks of the ranking (DeepConvShallowLSTM, Guan-LSTM, SA-HAR) lowers it to 0.36, and SVM and k-NN to 0.21; from the 13th network on, every

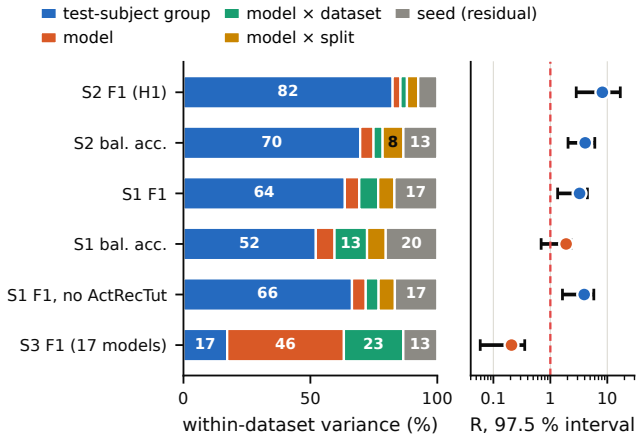


Fig. 4. Where the within-dataset variance of the stored runs lies. Left: shares by source (numbers in %; the seed term, grey, is the residual). Right: R with its 97.5 % bootstrap interval, blue where the interval lies above 1 and orange otherwise; the dashed line is the pre-registered threshold. R exceeds 1 in every analysis of the shared-skeleton networks (on S1 with balanced accuracy its interval includes 1), and the network dominates across the 17 heterogeneous published architectures (S3, conservative R'; by population in Fig. 5a).

interval lies below 1 (5 populations). The classical baselines alone do not explain the reversal: the 14 deep networks give $R' = 0.34$ [0.10, 0.64], and their largest term is the network × dataset interaction ($\sigma^2_{MD} = 56.4$, against $\sigma^2_M = 37.5$ and $\sigma^2_P = 42.7$; Fig. 5b). Across the published deep networks, which network suits which dataset outweighs both who is tested and which network is best overall.

D. Error budget of a comparison

This analysis is exploratory (r). With S2's components (Table 4, Fig. 5c), the true difference between two of its networks on a dataset has an SD of 4.2 F1 points. A comparison on one split and one seed has a noise SD of 6.1 points when both networks see the same test subjects, and of 17.1 when they do not, as when two studies each report one hold-out split: reliabilities of 0.32 and 0.06. The protocol's ten groups and three seeds, paired, bring the noise SD to 1.5 points (detectable difference 4.1, reliability 0.89): a typical difference between comparable networks lies at the limit of what one dataset can show. More seeds would remove at most 20 % of this noise, because the network × split interaction sets a floor of 1.2 points; twice as many test-subject groups would remove 29 %. A single network's dataset mean estimates its accuracy on new people of the same population with a standard error of 3.7 points, close to the typical difference between two networks. The development set S1 gives the same picture (Table 4).

E. Spread across test groups and networks

The decomposition's result is visible in the raw means (Fig. 6a). In 15 of 16 datasets of S1 and S2 the split means vary more than the network means, by a median factor of 2.6 in standard deviation; the exception is ActRecTut, whose two splits each test one of its two subjects. On UCI-HAR the split means have a standard deviation of 3.0 F1 points against 1.5 for the network means, and on UP-Fall 8.3 against 4.4. In the network × split cells (Fig. 6b, c) the columns

TABLE 4. Error budget of a comparison between two networks (F1 points)

	S2 (held-out)			S1 (development)		
Comparison	SD	Detect.	Rel.	SD	Detect.	Rel.
Unpaired, 1 × 1	17.1	47.9	0.06	13.5	37.7	0.13
Paired, 1 × 1	6.1	17.1	0.32	7.0	19.5	0.36
Paired, 10 × 1	1.9	5.4	0.82	2.2	6.2	0.85
Paired, 10 × 3 (protocol)	1.5	4.1	0.89	1.6	4.4	0.92
Paired, 10 × ∞	1.2	3.3	0.93	1.2	3.2	0.95
Paired, 20 × 3	1.0	2.9	0.94	1.1	3.1	0.96
True difference of two networks, SD	4.2			5.2		

Exploratory, from the point estimates of Table 3. Comparison: test-subject groups × seeds; paired: both networks on the same splits. SD: noise SD of the difference; Detect.: smallest true difference detected by a two-sided 5 % test with 80 % power; Rel.: reliability, true over true plus noise variance. Twenty groups means more recorded subjects.

dominate: a test group that is hard for one network is hard for the others.

F. Validation on the training subjects

Each run's validation windows are drawn, window by window, from its training subjects' recordings: they come from people the network has seen and can overlap its training windows, so the gap below mixes both effects [2] (Fig. 7). On the 13 datasets whose runs all recorded it, the validation macro-F1 exceeds the test macro-F1 in 2,307 of 2,388 runs, by a median of 17.1 points per dataset (from 4.6 to 47.8); the four largest gaps are on held-out datasets. The validation score is also a poor guide to the networks' test ranking, whereas the test ranking agrees with itself (r). The rank correlation between the networks' validation and test means has a median of +0.18 over the 13 datasets, against +0.80 between the test means of different seeds and +0.80 between those of different test subjects (even against odd splits); it is below the split-half value in 9 datasets. The network best on validation is the best on test in 3 of 13 datasets, the network best on the even splits the best on the odd ones in 7. On the odd splits, choosing by validation loses a median of 3.3 points, choosing by the even splits 0.0.

G. Exploratory findings

Within S1, the one design choice worth several points is temporal memory (Fig. 8a): S4D gains +5.79 [+2.30, +9.27] points over the no-memory control (9 of 9 datasets). This contrast changes more than the memory: the control is the shared skeleton with zero play widths ($D \times N = 32 \times 3$, 19,995 parameters on the reference input), whereas S4D uses its own, wider and shallower blocks (64×2, 27,675 parameters). The type of memory moves macro-F1 by at most a few points (Mamba − S4D: −2.45 [−4.40, −0.50]).

Over the 30 datasets, a larger channel count goes with a larger margin of the feature-based Random Forest over the best deep network (+7.1 points per standard deviation of $\log_2$ channels, other properties held fixed, Holm-adjusted $p < 0.001$; Fig. 8b). Of the 16 tests, 2 survive the correction: this one and the association of channel count with the spread over test-subject groups (adjusted $p = 0.025$). Fewer subjects go with a larger gap between the best and the median deep network (−1.5 per standard deviation, uncorrected $p = 0.004$), but not after the correction (adjusted $p = 0.058$); this gap, a maximum over

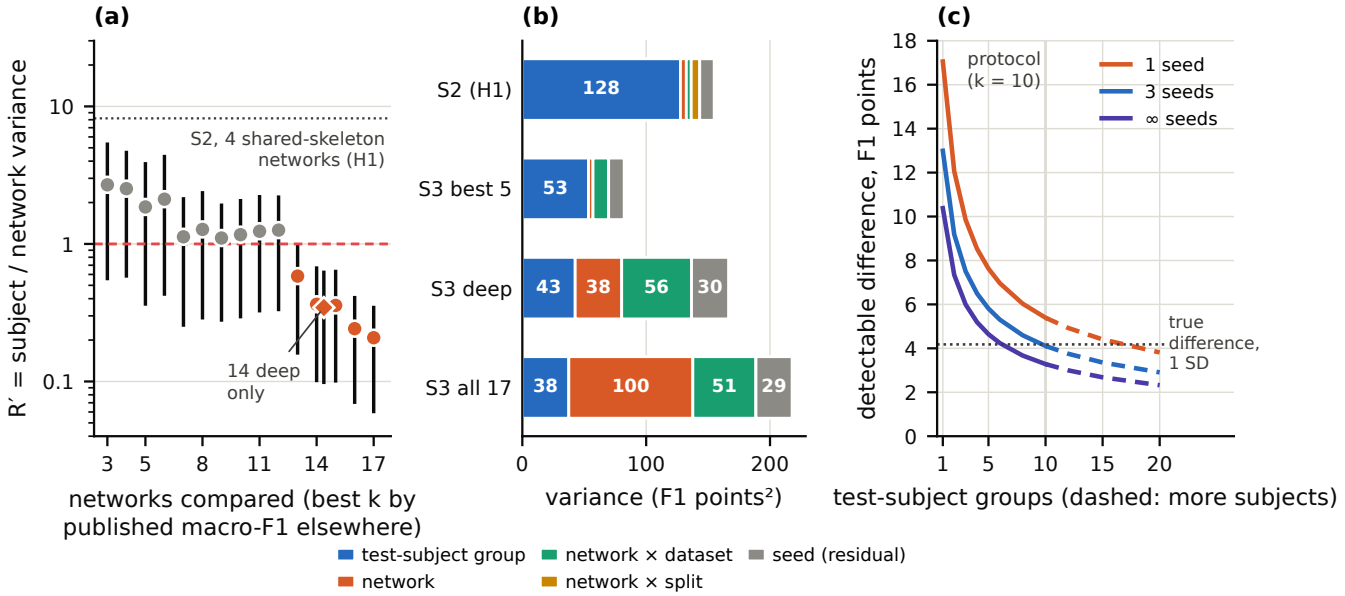


Fig. 5. Exploratory, added after review. (a) R' of S3 with its 97.5 % bootstrap interval for the k best of the benchmark's 17 networks, ranked by their mean published macro-F1 on the 26 Arena datasets outside S3 (circles), and for its 14 deep networks (diamond); blue where the interval lies above 1, orange below 1, grey across 1. Dotted: R of the pre-registered test (S2). (b) Variance components of S2 and of three S3 populations: the test-subject group dominates among comparable networks, the network × dataset interaction among the deep ones. (c) The smallest network difference a paired comparison detects (two-sided 5 % test, 80 % power) by number of test-subject groups and seeds, from S2's components; dashed beyond the protocol's ten groups (more subjects); dotted, the SD of the true difference between two networks. An unpaired single-split comparison would need 47.9 points.

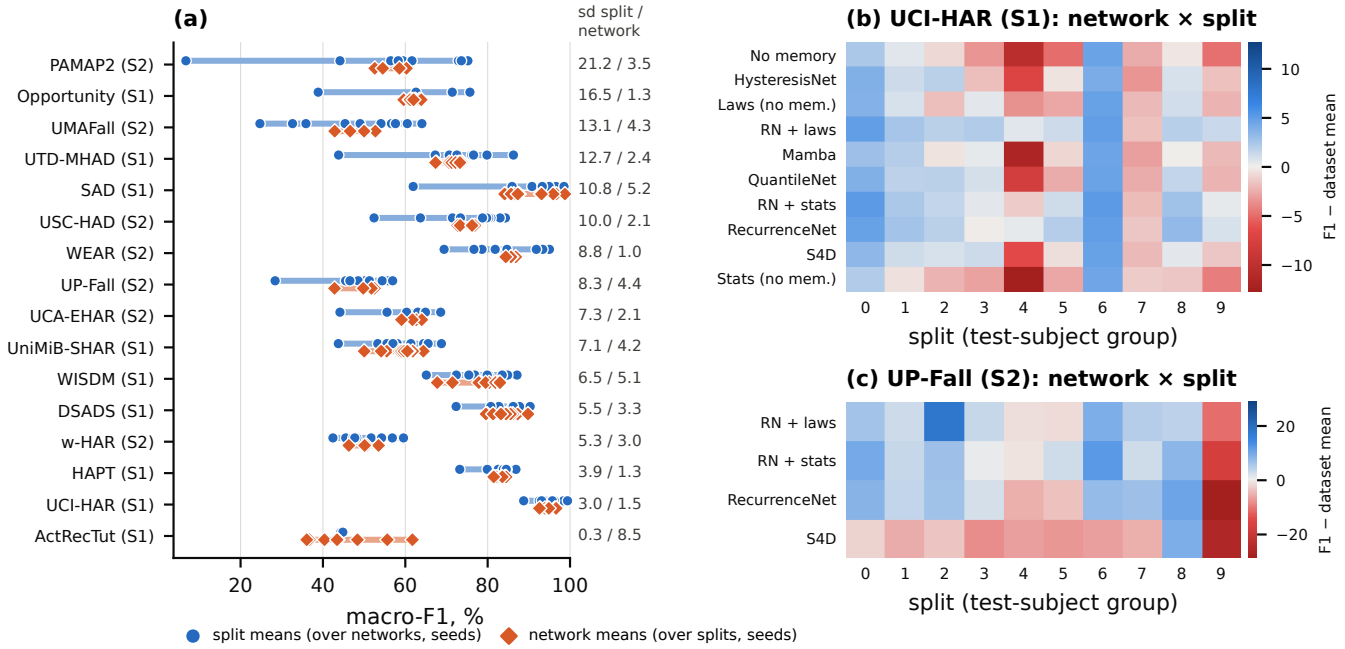


Fig. 6. Exploratory. (a) Within each dataset of S1 and S2, the means of the splits (circles; over networks and seeds) and of the networks (diamonds; over splits and seeds); right, the standard deviation of each. (b, c) Network × split cell means (over seeds) of UCI-HAR and UP-Fall, as deviations from the dataset mean.

noisy means, also rises with the spread over test groups ($\rho = +0.60$, $p < 0.001$), so it may reflect noisier estimates rather than larger network differences (r). No property predicts how hard a dataset is ($R^2 = 0.03$).

H. What the memories learned

The memories' own parameters barely move in training (Fig. 9; 766 checkpoints of eight S1 datasets, each against the same network freshly initialised). The quantile levels change by a median of 0.0047 (95th

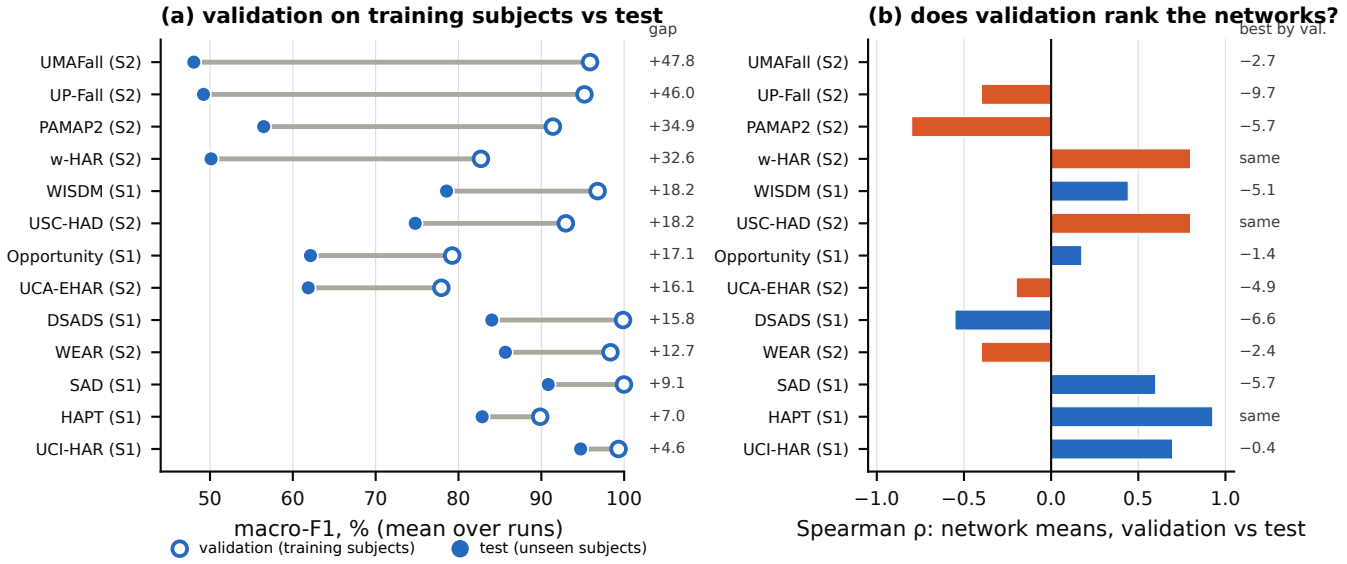


Fig. 7. Exploratory. (a) Per dataset, the mean validation macro-F1 (windows of the training subjects) and the mean test macro-F1 (unseen subjects) of the stored runs, and their gap. (b) Spearman correlation between the networks' validation and test means (S1 blue, S2 orange); right, whether the network best on validation is also best on test, or how many test points the choice loses. Datasets whose runs did not all record a validation score are omitted.

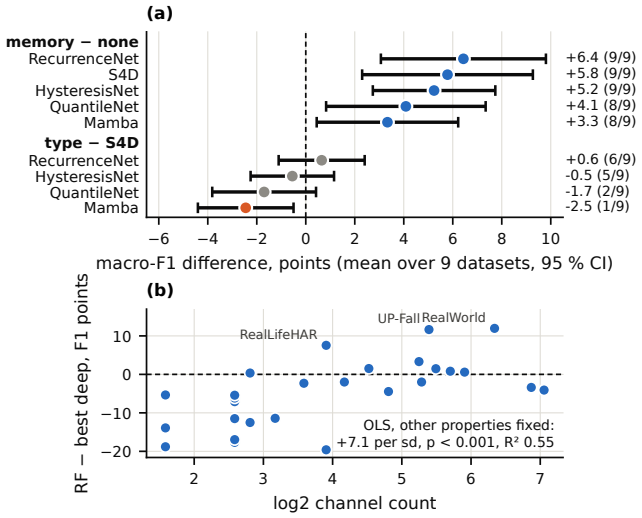


Fig. 8. Exploratory. (a) Temporal memory against none, and memory types against S4D, over the development datasets whose runs pair by subject (mean and 95 % CI, blue where the CI lies above 0, orange below 0, grey across 0; datasets won). (b) Over the benchmark's 30 datasets (published single runs), the Random Forest's margin over the best deep network rises with the number of sensor channels.

percentile 0.041), the recurrence radii by 0.014 (0.057), and S4D's decay rates by a median of 0.028 in $\log_{10}$. The play widths (median 0.36, initially 0.35) and S4D's mode time constants (median 1.8 window lengths, initially 1.6; 62 % longer than the window, initially 60 %) keep the distributions they start with. The time and amplitude scales at which each memory works are thus fixed by its design and initialisation; whatever memory adds (Fig. 8a), it adds at the scales it was given.

I. Embeddings

The hard test groups are the distant ones (Fig. 10). In a sensor-feature space that no network shapes (window statistics of the normalised signal), the splits whose test subjects lie further from the training subjects, class by class, are the splits on which the networks score lower: the pooled within-dataset rank correlation is $\rho = -0.32$ ($p = 0.002$), negative in 9 datasets. In the networks' embeddings the association is stronger, as it must partly be in the space the classifier reads: $\rho = -0.46$ ($p < 0.001$), negative in 10 of 10 datasets (median −0.61) and, for single networks, in 69 of 82 network–dataset pairs (median −0.50); per embedding dimension, so that networks of different width weigh alike, $\rho = -0.45$. The training embeddings still carry the subject's identity, as known for deep HAR features [31]: a linear probe identifies the training subject at a median 3.2 times chance, against 4.0 times from the window statistics, while the activity probe reaches a median balanced accuracy of 99 %. The networks thus reduce the subject information in their features without removing it. These associations are exploratory and correlational.

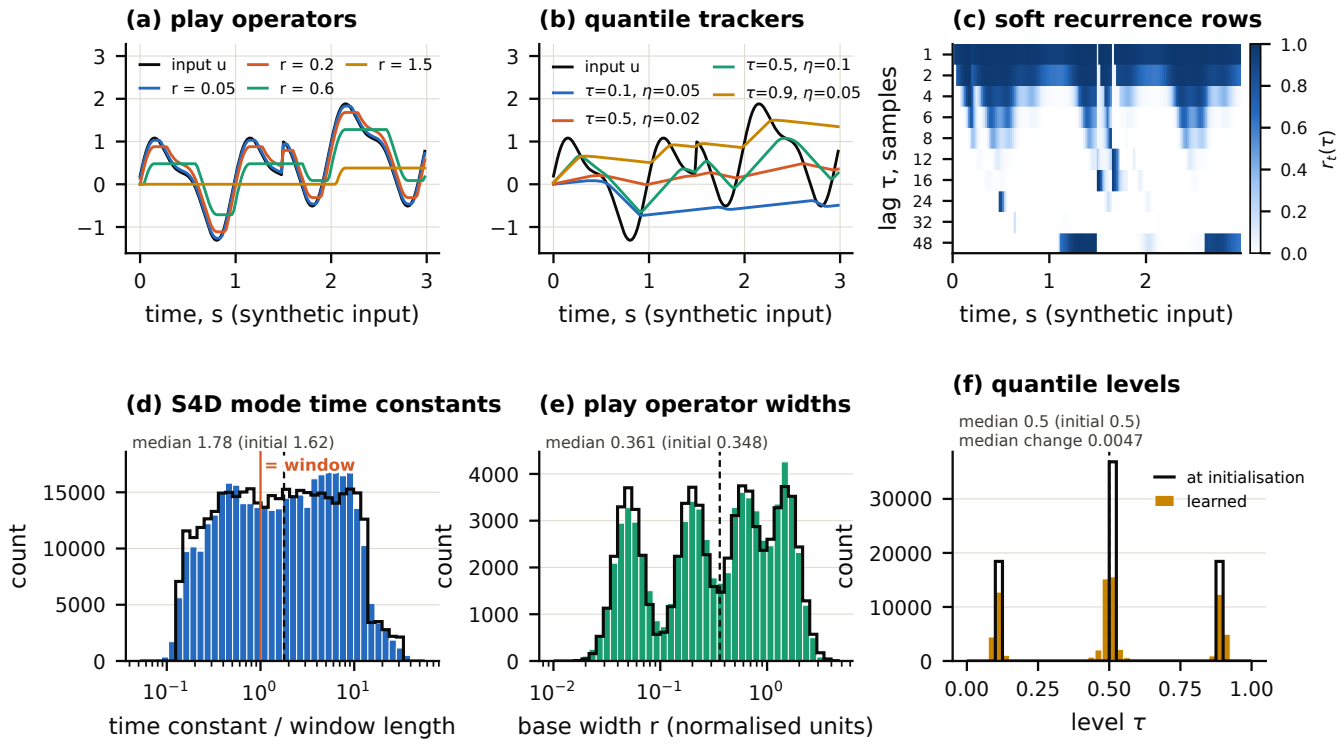


Fig. 9. The temporal memories. (a–c) The operators of HysteresisNet, QuantileNet and RecurrenceNet at their initial settings on a synthetic 3-s input (an illustration, not data). (d–f) Their parameters after training (bars) against the same networks freshly initialised (outline), over all stored checkpoints of eight S1 datasets: S4D's mode time constants in window lengths, the play widths, and the quantile levels.

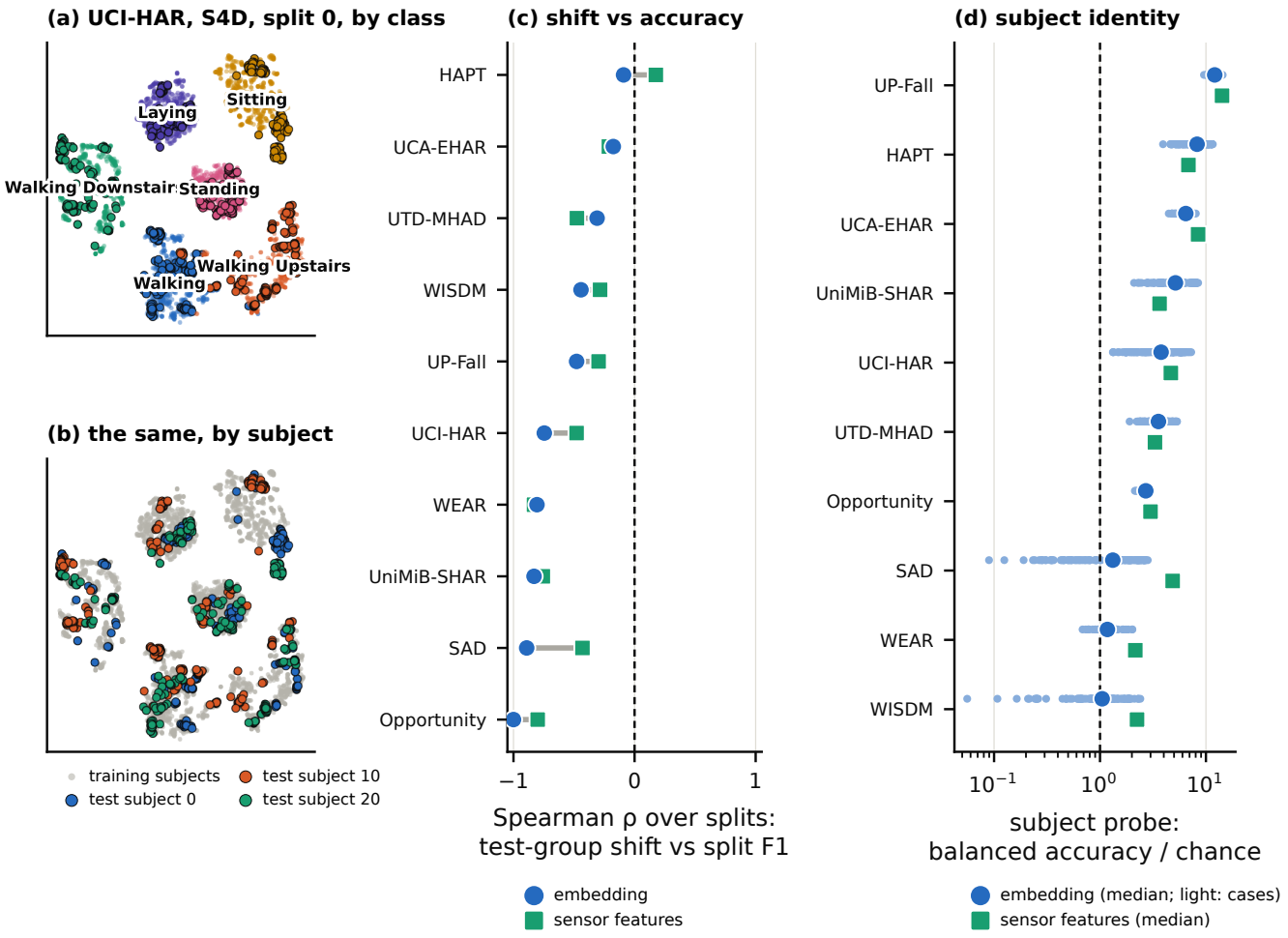


Fig. 10. Exploratory embedding analysis (seed-0 checkpoints, inference only). (a, b) t-SNE of the embeddings of one case fixed in advance (UCI-HAR, S4D, split 0): training windows (small, faded; a seeded subsample) and the test windows of the three unseen subjects (outlined), coloured by activity (a) and by subject (b). (c) Per dataset, Spearman correlation over splits between the test group's class-conditional shift (mean over networks) and the split's macro-F1, in the embedding and in a sensor-feature space. (d) Balanced accuracy of a linear probe that identifies the training subject, over chance, from the embeddings (all cases and median) and from the window statistics (median).

IV. DISCUSSION AND LIMITATIONS

What is new. That accuracy varies widely between people was known [6], [7]; how that variation compares with the differences between networks, and what it implies for a comparison, was not. The decomposition answers both under one protocol. For networks that differ in single components, so that their dataset means differ by a few points, who is tested decides more of a reported macro-F1 than which network is used, as the pre-registered test confirmed on held-out data. The ratio is not a constant of wearable HAR but a property of the networks compared: it holds near or above 1 among the benchmark's best published networks and reverses once weaker ones enter, and among the published deep networks the largest term is the network × dataset interaction, a quantitative form of the benchmark's finding that no architecture leads everywhere [9].

What it implies for a study. The variance between test-subject groups governs the absolute level, and with it every comparison made across studies, splits or subject groups: an unpaired comparison on one hold-out split has a reliability of 0.06 (Table 4). A comparison paired by split cancels it, but is then limited by the network × split interaction and the seed, which on S2 are as large as the network terms themselves (Table 3). Hence four rules for a wearable sensing study. Compare networks paired on the same splits, never by means over different subjects. Size the comparison by test subjects: with the protocol's ten groups and three seeds, about 4.1 F1 points are detectable on one dataset, and further seeds cannot go below the floor that the network × split interaction sets. Select networks and stop training on held-out subjects, since validation windows of the training subjects rank networks far worse than the test ranks them against itself (Section III-F). Report the spread across test-subject groups, and the standard error of a mean as an estimate for new people (3.7 points on S2), with every score.

Exploratory evidence. Within datasets, test groups spread accuracy more than networks do (Fig. 6). A test group is hard when its windows, class by class, lie far from the training subjects', already in features that no network shapes, and the networks' embeddings still identify the training subjects (Fig. 10), whereas the temporal memories keep the scales they start with (Fig. 9). Together these suggest that, for comparable networks, the training subjects and how representations align them are more promising levers for accuracy on new people than another temporal operator. Over the benchmark's 30 datasets, the one association that survives correction and concerns the comparison of models points at the sensing system: with many channels, hand-crafted features stay competitive.

Limitations. The analysis covers one protocol, and three seeds per run. The S1 and S2 networks share one skeleton, so their components are controlled but not independent designs. S2's means and paired network differences were known before the registration. Only H1 was pre-registered: the network-population, error-budget and validation analyses were added after an internal review and are exploratory. S3 has one run per split on four datasets, so its seed and interaction terms are merged. The error budget uses point estimates, and σ^2_M from four networks is imprecise. P is the effect of the subject partition, mainly but not only of the test group. The 30-dataset results re-analyse published single runs and are correlational, and subjects and channels co-vary across datasets. The embedding analysis uses the seed-0 checkpoints of eleven datasets, and the validation analysis only the datasets whose runs recorded a validation score; on S2, with four networks, its rank correlations are coarse. The intervals resample test-subject groups and treat the networks as fixed: they say how precisely R is known for these networks, not for networks in general, and the S3 populations show how it changes with the networks compared.

V. CONCLUSION

Under a fixed person-disjoint protocol, who is tested accounts for most of the variance of macro-F1 among comparable wearable HAR networks, as a pre-registered test on held-out datasets confirmed; among the benchmark's best published networks the two are of similar size, and once weaker networks enter, the network dominates. The variance components turn this into an error budget: comparisons must be paired by split, sized by test subjects rather than seeds, and selected on held-out subjects, and studies of wearable sensing systems should report the spread across their test subjects.

REFERENCES

[1] A. Bulling, U. Blanke, and B. Schiele, "A tutorial on human activity recognition using body-worn inertial sensors," *ACM Computing Surveys*, vol. 46, no. 3, pp. 33:1–33:33, 2014.

[2] N. Y. Hammerla and T. Plötz, "Let's (not) stick together: Pairwise similarity biases cross-validation in activity recognition," in *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. Osaka, Japan: ACM, 2015, pp. 1041–1051.

[3] J. W. Lockhart and G. M. Weiss, "Limitations with activity recognition methodology & data sets," in *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication (UbiComp '14 Adjunct)*. Seattle, WA, USA: ACM, 2014, pp. 747–756.

[4] H. Qian, S. J. Pan, and C. Miao, "Latent independent excitation for generalizable sensor-based cross-person activity recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 13, 2021.

[5] A. Stisen, H. Blunck, S. Bhattacharya, T. S. Prentow, M. B. Kjærgaard, A. Dey, T. Sonne, and M. M. Jensen, "Smart devices are different: Assessing and mitigating mobile sensing heterogeneities for activity recognition," in *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems (SenSys '15)*. Seoul, South Korea: ACM, 2015, pp. 127–140.

[6] B. Barshan and A. Yurtman, "Investigating inter-subject and inter-activity variations in activity recognition using wearable motion sensors," *The Computer Journal*, vol. 59, no. 9, pp. 1345–1362, 2016.

[7] A. Jordao, A. C. Nazare, J. Sena, and W. R. Schwartz, "Human activity recognition based on wearable sensor data: A standardization of the state-of-the-art," arXiv:1806.05226, 2018.

[8] G. Varoquaux, "Cross-validation failure: Small sample sizes lead to large error bars," *NeuroImage*, vol. 180, pp. 68–77, 2018.

[9] M. Burzer, T. King, T. Riedel, M. Beigl, and T. Röddiger, "WHAR Arena: Benchmarking the state of the art in efficient wearable human activity recognition," arXiv:2606.13194, 2026.

[10] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Sepah, E. Raff, K. Madan, V. Voleti, S. E. Kahou, V. Michalski, D. Serdyuk, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 3, 2021.

[11] M. Burzer, T. King, T. Riedel, M. Beigl, and T. Röddiger, "WHAR datasets: An open source library for wearable human activity recognition," in *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*. Espoo, Finland: ACM, 2025, pp. 1315–1322.

[12] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *International Conference on Learning Representations (ICLR)*, 2022.

[13] A. Gu, A. Gupta, K. Goel, and C. Ré, "On the parameterization and initialization of diagonal state space models," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[14] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv:2312.00752, 2023.

[15] F. J. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.

[16] N. Y. Hammerla, S. Halloran, and T. Plötz, "Deep, convolutional, and recurrent models for human activity recognition using wearables," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2016.

[17] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[18] M. Christ, N. Braun, J. Neuffer, and A. W. Kempa-Liehr, "Time series FeatuRe extraction on basis of scalable hypothesis tests (tsfresh – A Python package)," *Neurocomputing*, vol. 307, pp. 72–77, 2018.

[19] J.-P. Eckmann, S. O. Kamphorst, and D. Ruelle, "Recurrence plots of dynamical systems," *Europhysics Letters*, vol. 4, no. 9, pp. 973–977, 1987.

[20] M. Brokate and J. Sprekels, *Hysteresis and Phase Transitions*, ser. Applied Mathematical Sciences. New York: Springer, 1996, vol. 121.

[21] H. Robbins and S. Monro, "A stochastic approximation method," *The Annals of Mathematical Statistics*, vol. 22, no. 3, pp. 400–407, 1951.

[22] H. D. Patterson and R. Thompson, "Recovery of inter-block information when block sizes are unequal," *Biometrika*, vol. 58, no. 3, pp. 545–554, 1971.

[23] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with Python," in *Proceedings of the 9th Python in Science Conference (SciPy 2010)*, Austin, TX, USA, 2010, pp. 92–96.

[24] C. A. Field and A. H. Welsh, "Bootstrapping clustered data," *Journal of the Royal Statistical Society Series B*, vol. 69, no. 3, pp. 369–390, 2007.

[25] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[26] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006. [Online]. Available: https://www.jmlr.org/papers/v7/demsar06a.html

[27] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[28] K. Zhang, B. Schölkopf, K. Muandet, and Z. Wang, "Domain adaptation under target and conditional shift," in *Proceedings of the 30th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 28, no. 3, 2013, pp. 819–827. [Online]. Available: https://proceedings.mlr.press/v28/zhang13d.html

[29] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," arXiv preprint arXiv:1610.01644, 2016.

[30] L. van der Maaten and G. Hinton, "Visualizing data using t-sne," *Journal of Machine Learning Research*, vol. 9, no. 86, pp. 2579–2605, 2008. [Online]. Available: https://www.jmlr.org/papers/v9/vandermaaten08a.html

[31] Y. Iwasawa, K. Nakayama, I. Yairi, and Y. Matsuo, "Privacy issues regarding the application of DNNs to activity-recognition using wearables and its countermeasures by use of adversarial training," in *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*, 2017, pp. 1930–1936.